

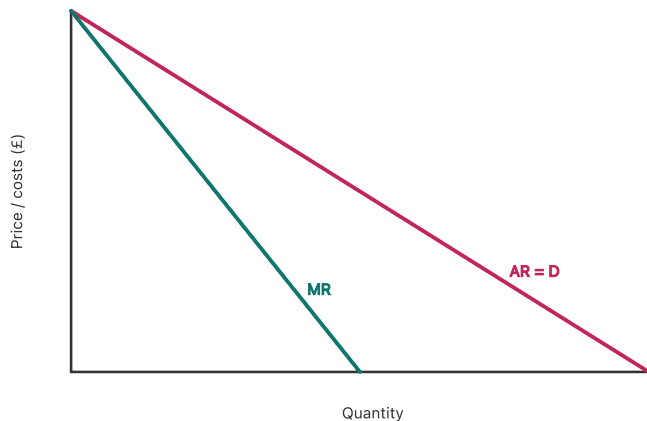
# Chapter 18: Revenues, costs, profits and objectives

Edexcel A-Level Economics, Theme 3 (spec 3.2.1 and 3.3.1 to 3.3.4) · Class notes

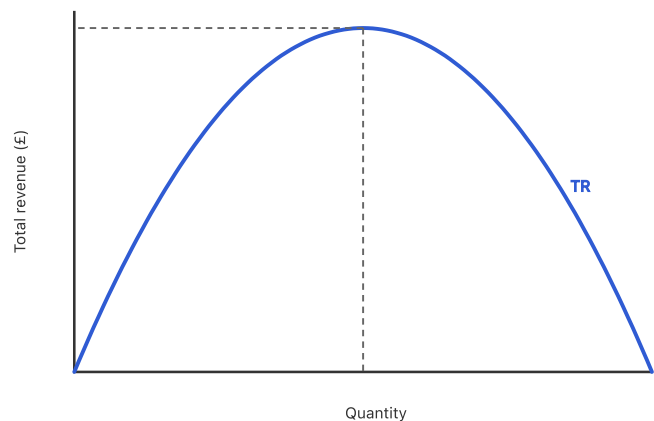
## 1. Revenue

$$TR = P \times Q \quad AR = TR \div Q = P \quad MR = \text{change in TR} \div \text{change in Q}$$

AR is the firm's demand curve. When price must fall to sell more, MR falls faster than AR because the lower price applies to every unit, so MR lies below AR (twice as steep for a straight-line demand curve). A price taker (perfect competition) has  $AR = MR = P$ .



$$AR = 40 - 2Q \text{ and } MR = 40 - 4Q$$



$$TR \text{ peaks where } MR = 0 \text{ (} Q = 10 \text{)}$$

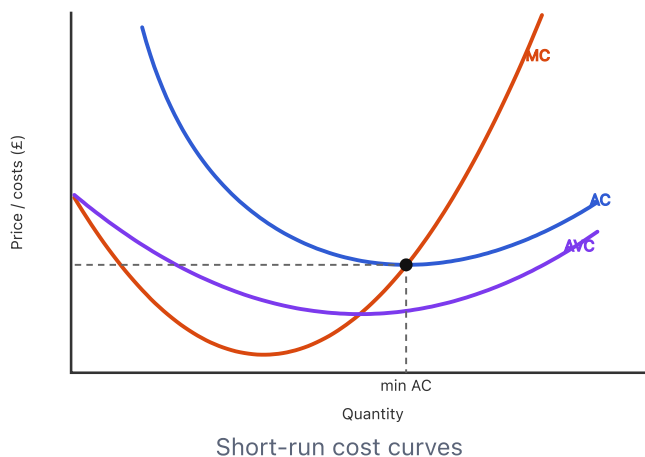
- $MR > 0$ : demand is elastic ( $PED > 1$ ); cutting price raises TR.
- $MR = 0$ : unit elastic; TR is at its maximum.
- $MR < 0$ : inelastic ( $PED < 1$ ); cutting price lowers TR.

## 2. Costs

$$TC = TFC + TVC \quad AC = TC \div Q \quad AVC = TVC \div Q \quad AFC = TFC \div Q \quad MC = \text{change in TC} \div \text{change in Q}$$

Short run: at least one factor is fixed. Long run: all factors are variable. Fixed costs (rent, insurance) do not change with output; variable costs (materials, hourly wages) do.

**Law of diminishing marginal productivity:** with a fixed factor, extra workers first add more and more to output (specialisation), then less and less (crowding). Marginal product falls, so MC rises, giving the U shape.

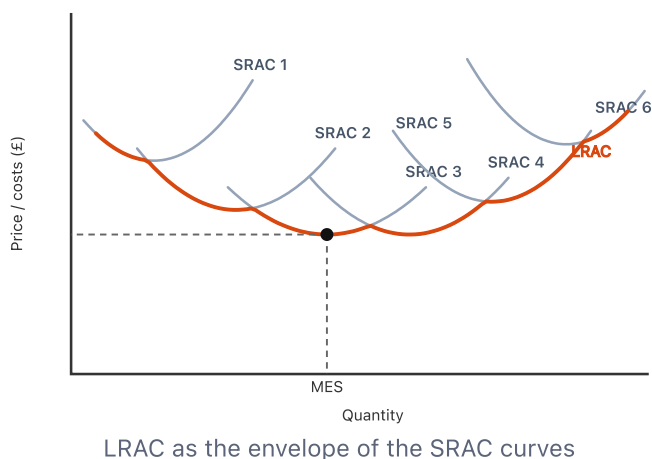


### Reading the diagram

- MC cuts AVC and AC at their minimum points.
- Below AC, MC pulls the average down; above it, MC pulls it up.
- $AC = AVC + AFC$ . The gap between AC and AVC is AFC, which shrinks as output rises.

## 3. Long-run costs, economies and diseconomies of scale

The LRAC is the lower envelope of all the SRAC curves: the cheapest cost of each output when every factor can change. The minimum efficient scale (MES) is the lowest output at which LRAC is at its minimum.



### Internal economies

- Technical: large machinery, increased dimensions
- Purchasing: bulk-buy discounts
- Managerial: specialist managers
- Financial: cheaper borrowing
- Marketing: costs spread over more units
- Risk-bearing: diversification

### External economies

Supplier clusters, skilled labour pool, shared infrastructure.

### Diseconomies of scale

Internal: communication, coordination, motivation and control problems. External: congestion, rising rents and wages.

#### Exam tip

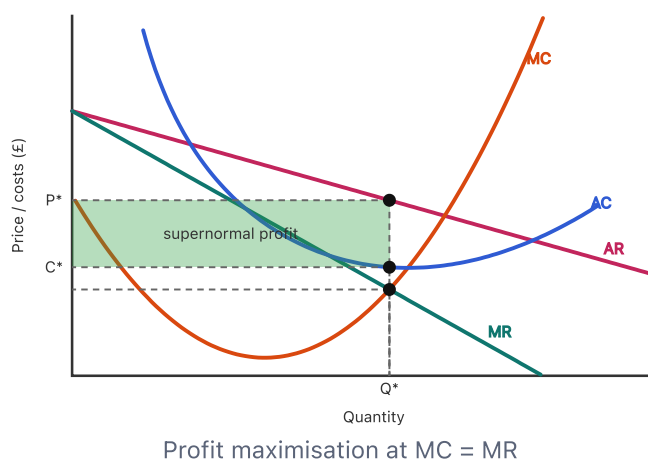
Pick the economy that fits the business in the case, then chain it: bulk buying, so lower cost per unit, so lower LRAC, so the firm can cut price or widen its margin.

## 4. Profit

$$\text{Profit} = TR - TC \quad \text{Profit per unit} = AR - AC$$

- **Normal profit:** the minimum reward to keep the entrepreneur in the industry (an opportunity cost inside AC).  $AR = AC$ .
- **Supernormal profit:**  $AR > AC$ . Attracts entry in competitive markets.

- **Loss:**  $AR < AC$ .



### Profit maximisation

1. Find  $Q^*$  where  $MC = MR$ .
2. Go up to  $AR$  to read  $P^*$ .
3. Go across from  $AC$  at  $Q^*$  to read  $C^*$ .
4. Profit =  $(P^* - C^*) \times Q^*$ , the shaded rectangle.

### Shut-down points

- Short run: shut down if  $AR < AVC$ .
- Long run: exit if  $AR < AC$ .

## 5. Business objectives

| Objective            | Rule                                       | Compared with profit max                   |
|----------------------|--------------------------------------------|--------------------------------------------|
| Profit maximisation  | $MC = MR$                                  | Baseline                                   |
| Revenue maximisation | $MR = 0$                                   | Higher output, lower price                 |
| Sales maximisation   | $AC = AR$                                  | Highest output with normal profit, no loss |
| Satisficing          | Enough profit to keep shareholders content | Below max profit                           |

Managers may prefer revenue or sales because pay and status rise with firm size, while shareholders want profit: the principal-agent problem (Chapter 17).

### Key terms

**Total revenue** price  $\times$  quantity. **Average revenue**  $TR \div Q$ , the demand curve. **Marginal revenue** extra revenue from one more unit. **Marginal cost** extra cost of one more unit. **Diminishing marginal productivity** extra workers add less output once a factor is fixed. **Economies of scale** LRAC falls as output rises. **Diseconomies of scale** LRAC rises as output rises. **MES** lowest output at minimum LRAC. **Normal profit**  $AR = AC$ . **Supernormal profit**  $AR > AC$ . **Satisficing** a satisfactory, not maximum, profit.

### Practice

1. Explain why  $MR$  lies below  $AR$  for a firm facing a downward-sloping demand curve. (4)
2. Using a diagram, explain how a firm finds its profit-maximising output and the profit it makes. (6)
3. Explain one internal and one external economy of scale. (4)
4. Evaluate the view that a firm should always aim to maximise profit. (25)